

1. Replacement of AlexNet with custom CNN.

The baseline 2016 SER paper deployed LSTM on top of AlexNet module which was trained on classifying images into classifiers such as dog,cat,human,etc. Although it was a well trained model and only had to be fine tuned for speech it was not necessary the best for Extracting Features from Mel Spectrogram because it wasn't designed to work on audio hence we came across a reserch paper implement a 3 way CNN filter For mel spectrogram they did a combined CNN approach while taking both Audio and Mel spectrogram as their input. And using 32,64,128 filters as their base which became the baseline for our CNN pipeline.

<https://www.mdpi.com/1424-8220/24/17/5797>

Although Dual CNN approach wasn't applied to this project due to computational requirements of it and thus the real world boundations although it may be implemented to boost F1 scores and Accuracies even further for working on all Datasets.

We got the basic idea to use 32,64,128 filters as the starting point from here and

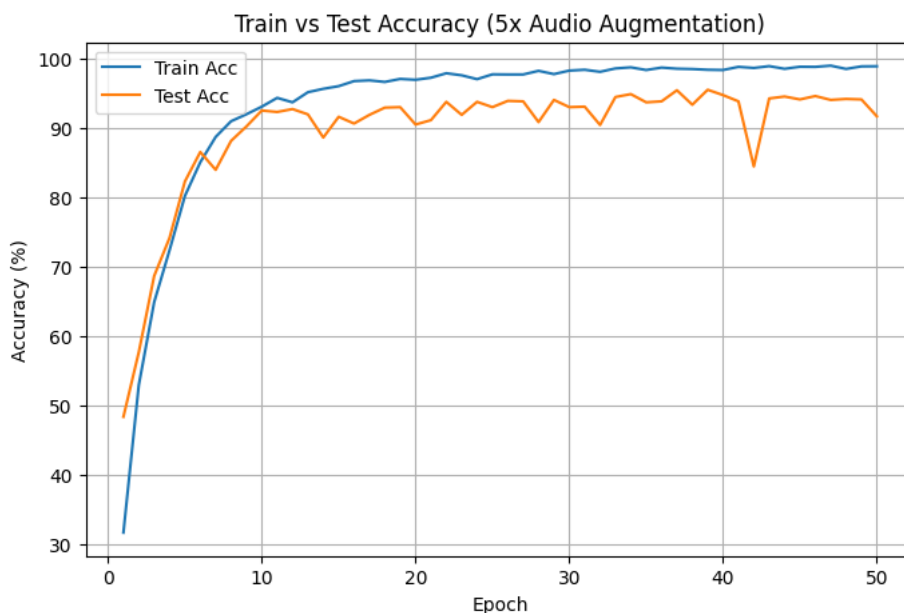
2. Data Augmentation 5x RAVDESS AUGMENTATION

<https://ieeexplore.ieee.org/document/10392486> (Paper 1)

<https://www.mdpi.com/2078-2489/16/7/518>

This reserch paper(PAPER 1) showed a 30 percent boost in accuracy after data augmentation on the DATASET performing similar Augmentation of Gaussian Noise,Time Shifting,Pitch Shifting,Time stretch and Super Perturb to the original audios before generating mel spectrogram for the pipeline boosted our accuracy from 88 percent to 95.49 or 97% through expanding the dataset to 5x audio spaces,i.e 7200 as compared to 1200. The result log is given below

Epoch 39/50 | Train Acc: 98.37% | Test Acc: 95.49%



3. BLSTM + ATTENTION MODULE .

Original Paper Stated the use of attention + BLSTM where BLSTM Helped Find the Dependence of emotion on time and further Attention module finds the time frames which are most important for a smoother classifying process and only important frames being noticed.

A very basic BLSTM + ATTENTION module is used here with the main focus towards GWO optimization

4. GWO Optimizer for Meta Heuristic Training of Model

<https://www.mdpi.com/2673-4591/59/1/214>

GWO optimizer idea was adopted from the above Paper where when applied it worked way better than genetic algorithm for low computation the wolves in the above paper selected which EEG (electroencephalogram) features to keep and which ones to discard while Analysing the accuracy on binary operations of removing adding one of the 8 features this worked on the brain Impulses which is way different than our speech approach also instead of selecting which features to keep our model focuses more on modifying the whole pipeline using this meta heuristic approach pre training in order to obtain best close Architecture without overfitting the model which is evident through difference in train and test accuracies.

Our model works on the following parameters hyperparameter tuning .

1-**CNN FILTER SIZES**- The original 32,64,128 filter size might not be the optimal choice for all datasets and robust fitness thus we use GWO to extract filter sizes which works best on our dataset thus finding the closest possible architecture for certain datasets on training on larger datasets with similar approach we can reach to the closest to speech recognition architecture which works best with all types of human voices.

2- **Dropout Rate** – This prevents overfitting by randomly turning off certain neurons in CNN models and thus its Percentage defines the overall fitness of the model.

3- **lstm_hidden** – this tells the capacity of BLSTM + ATTENTION module to keep the relevant data basically a lstm of 128 means it has 128 spaces for each time stamp of forward and backward thus a total of 256 memory array for each time stamp.

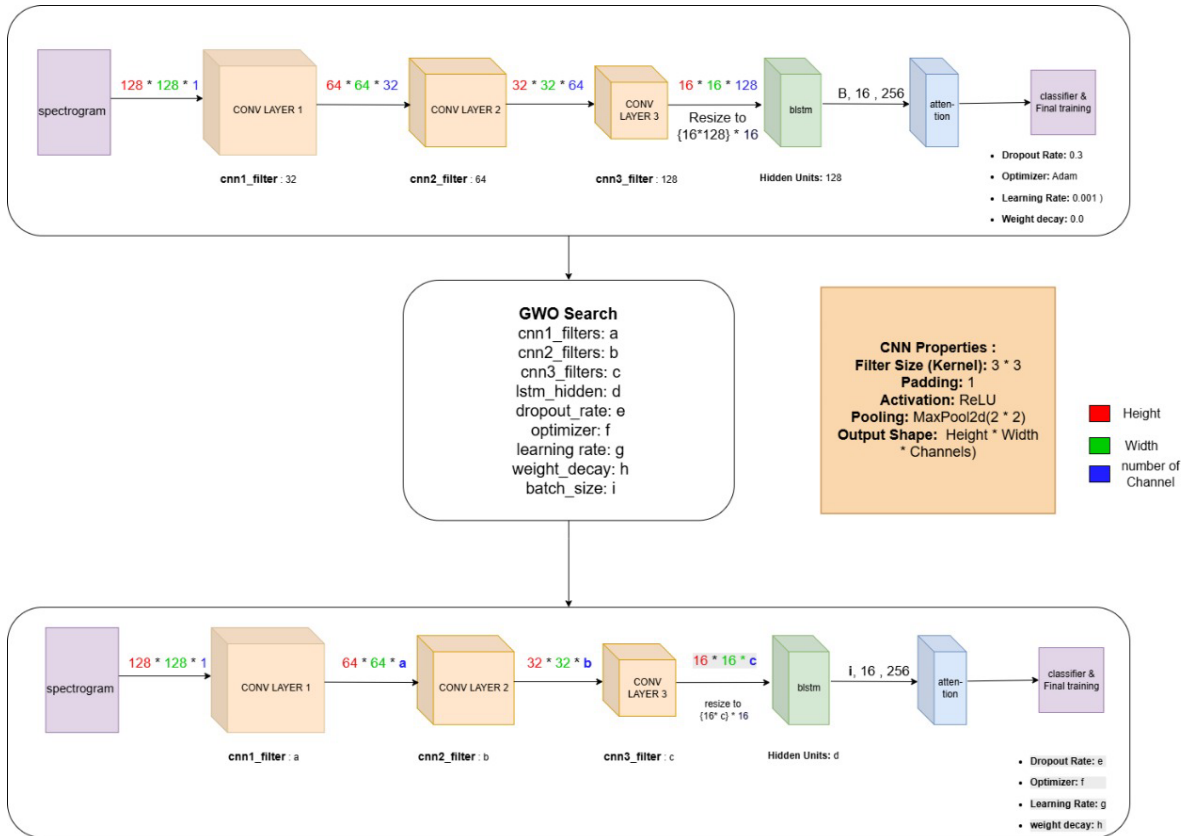
4- **Optimizer** – used for gradient descent in the final model training(this thus not change the architecture rather only updates the matrices values of filters etc).

5- **Learning Rate** – This affects how much the values of gradient descent in the final training change on each step.

6-**Weight_decay**- Penalty on Very high weighted values to promote generalization on the final training and avoid overfitting as well.

Model And GWO Architectures are given below:-

Our Approach and Architecture



GWO OPTIMIZER ROLE:-

